

Pharm LMS — EC2 Deployment Guide

Audience: Developer deploying Pharm LMS on AWS EC2

Strategy: Clone from your personal repo; push updates to **both** your repo and the company repo

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Overview

Piece	Choice
Server	AWS EC2 t3.micro (Amazon Linux 2023)
Clone source (EC2)	<code>https://github.com/Prince-213/pharm-lms.git</code>
Push targets (your laptop)	<code>origin</code> (personal) + <code>company</code> (PharmAnalytics)
Database	Neon PostgreSQL (<code>DATABASE_URL</code>)
File storage	Cloudflare R2
Process manager	PM2
Web proxy	Nginx (ports 80 / 443)

EC2 does **not** auto-deploy when you push to GitHub. After each push, SSH to the server and run `~/deploy-pharm-lms.sh`.

Part 1 — Git setup on your laptop (dual push)

Your machine already has two remotes:

Remote	Repository
<code>origin</code>	<code>https://github.com/Prince-213/pharm-lms.git</code>
<code>company</code>	<code>https://github.com/PharmAnalytics/PharmAnalytics-LMS-.git</code>

Push to both repos after every release

```
git add .  
git commit -m "your message"  
git push origin main  
git push company main
```

Optional: one command alias

```
git config alias.pushall "!git push origin main && git push company main"
```

Then: `git pushall`

Rule: Keep both repos on the same `main` commit so the company always has your latest code, even though the server clones from **your** repo.

Part 2 — Create the EC2 instance

1. AWS Console → **EC2** → **Launch instance**
2. **Name:** `pharm-lms`
3. **AMI:** Amazon Linux 2023
4. **Instance type:** `t3.micro` (free tier)
5. **Key pair:** Create or use existing (or use EC2 Instance Connect without PEM)
6. **Network settings — check both:**
 - Allow **HTTP** traffic from the internet (port 80)
 - Allow **HTTPS** traffic from the internet (port 443)
7. **Storage:** 20–30 GB
8. **Launch instance**

Elastic IP (recommended)

1. EC2 → **Elastic IPs** → **Allocate**
2. **Associate** with your `pharm-lms` instance
3. Note the IP as **YOUR_IP** (e.g. `3.15.xxx.xxx`)

Budget alert

AWS Billing → Budgets → alert at **\$10** and **\$25** (protect your \$100 credit).

Part 3 — Connect to the server

Option A — EC2 Instance Connect (no PEM)

1. EC2 → Instances → select instance → **Connect**
2. Tab: **EC2 Instance Connect**
3. User: `ec2-user`
4. **Connect**

Option B — SSH with key

```
ssh -i "path\to\your-key.pem" ec2-user@YOUR_IP
```

Part 4 — First-time server setup

Run all commands on the EC2 instance.

4.1 System packages

```
sudo dnf update -y
sudo dnf install -y git nginx
```

4.2 Node.js 22 + pnpm

```
curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.40.1/install.sh | bash
source ~/.bashrc
nvm install 22
```

```
nvm use 22
```

```
nvm alias default 22
```

```
corepack enable
```

```
corepack prepare pnpm@9.15.9 --activate
```

```
node -v
```

```
pnpm -v</pre>
```

4.3 Swap (required on 1 GB RAM for `pnpm build`)

```
<pre class="code-block">sudo fallocate -l 2G /swapfile
sudo chmod 600 /swapfile
sudo mkswap /swapfile
sudo swapon /swapfile
echo '/swapfile none swap sw 0 0' | sudo tee -a /etc/fstab
free -h</pre>
```

4.4 PM2

```
<pre class="code-block">pnpm add -g pm2</pre>
```

Part 5 — Clone your personal repo on EC2

Clone from **Prince-213/pharm-lms** (not the company repo — avoids org token issues).

If the repo is public

```
<pre class="code-block">cd ~
git clone https://github.com/Prince-213/pharm-lms.git pharm-lms
cd pharm-lms</pre>
```

If the repo is private

Option A — Personal Access Token (HTTPS)

1. GitHub → Settings → Developer settings → Personal access tokens
2. Create token with **Contents: Read** on Prince-213/pharm-lms
3. Clone:

```
<pre class="code-block">git clone https://github.com/Prince-213/pharm-lms.git pharm-lms</pre>
```

- Username: Prince-213
- Password: paste the **token** (not your GitHub password)

Option B — Deploy key (SSH)

```
<pre class="code-block">ssh-keygen -t ed25519 -C "pharm-lms-ec2" -f ~/.ssh/pharm_lms_deploy -N ""  
cat ~/.ssh/pharm_lms_deploy.pub</pre>
```

Add the public key in **your** repo: Settings → Deploy keys → Add deploy key.

```
<pre class="code-block">nano ~/.ssh/config</pre>
```

```
<pre class="code-block">Host github.com  
HostName github.com  
User git  
IdentityFile ~/.ssh/pharm_lms_deploy  
IdentitiesOnly yes</pre>
```

```
<pre class="code-block">chmod 600 ~/.ssh/config ~/.ssh/pharm_lms_deploy  
git clone git@github.com:Prince-213/pharm-lms.git pharm-lms  
cd pharm-lms</pre>
```

Part 6 — Production environment file

```
<pre class="code-block">cd ~/pharm-lms  
nano .env</pre>
```

Paste production values. **Minimum required:**

```
<pre class="code-block">DATABASE_URL=postgresql://USER:PASSWORD@HOST/neondb?sslmode=require  
AUTH_SECRET=long-random-secret-here</pre>
```

AUTH_TRUST_HOST=true

AUTH_URL=http://YOUR_IP

NEXTAUTH_URL=http://YOUR_IP

NEXT_PUBLIC_SITE_URL=http://YOUR_IP

R2_ENDPOINT=https://....r2.cloudflarestorage.com

R2_REGION=auto

R2_ACCESS_KEY_ID=...

R2_SECRET_ACCESS_KEY=...

R2_BUCKET_NAME=pharm-lms-files

R2_CORS_ORIGINS=http://YOUR_IP,http://localhost:3000

RESEND_API_KEY=...

EMAIL_FROM=Pharm LMS <noreply@yourdomain.com>

PAYSTACK_SECRET_KEY=...

NEXT_PUBLIC_PAYSTACK_PUBLIC_KEY=...

AUTH_GOOGLE_ID=...

AUTH_GOOGLE_SECRET=...

PG_POOL_MAX=5</pre>

Save: `Ctrl+O` , Enter, `Ctrl+X` .

Never commit `.env` to GitHub.

When you add a domain and HTTPS, change the three URL variables to `https://yourdomain.com` and run `pm2 restart pharm-lms` .

Part 7 — First deploy (build and start)

```
cd ~/pharm-lms
```

```
pnpm install --frozen-lockfile
```

```
pnpm db:migrate:deploy
```

```
pnpm build
```

```
pm2 start "pnpm exec next start" --name pharm-lms
```

```
pm2 save
```

```
pm2 startup</pre>
```

Run the `sudo env PATH=...` command that `pm2 startup` prints, then:

```
<pre class="code-block">pm2 save</pre>
```

Verify:

```
<pre class="code-block">curl -I http://127.0.0.1:3000
pm2 logs pharm-lms --lines 30</pre>
```

Part 8 — Nginx (public HTTP)

```
<pre class="code-block">sudo nano /etc/nginx/conf.d/pharm-lms.conf</pre>
```

```
<pre class="code-block">server {
listen 80;
server_name _;

location / {
proxy_pass http://127.0.0.1:3000;
proxy_http_version 1.1;
proxy_set_header Upgrade $http_upgrade;
proxy_set_header Connection 'upgrade';
proxy_set_header Host $host;
proxy_set_header X-Real-IP $remote_addr;
proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
proxy_set_header X-Forwarded-Proto $scheme;
proxy_cache_bypass $http_upgrade;
}
}</pre>
```

```
<pre class="code-block">sudo nginx -t
sudo systemctl enable nginx
sudo systemctl restart nginx</pre>
```

Open in browser: **`http://YOUR_IP`**

Part 9 — Install the deploy script (one-time)

The repo includes `scripts/deploy-pharm-lms.sh` . Copy it to your home directory:

```
<pre class="code-block">cp ~/pharm-lms/scripts/deploy-pharm-lms.sh ~/deploy-pharm-lms.sh
chmod +x ~/deploy-pharm-lms.sh</pre>
```

Part 10 — Update the live site (every release)

On your laptop

1. Make code changes
2. Commit and push to **both** remotes:

```
<pre class="code-block">git add .
git commit -m "describe your change"
git push origin main
git push company main</pre>
```

On EC2

1. Connect (Instance Connect or SSH)
2. Run:

```
<pre class="code-block">~/deploy-pharm-lms.sh</pre>
```

That script will:

1. `git pull origin main` from `~/pharm-lms`
2. `pnpm install --frozen-lockfile`
3. `pnpm db:migrate:deploy`
4. `pnpm build`
5. `pm2 restart pharm-lms`

Verify

```
<pre class="code-block">pm2 status  
pm2 logs pharm-lms --lines 50</pre>
```

Refresh the site in your browser.

When you only changed `.env`

```
<pre class="code-block">nano ~/pharm-lms/.env  
pm2 restart pharm-lms</pre>
```

No need to pull or build.

Part 11 — HTTPS and custom domain (optional)

1. Point DNS **A record** → YOUR_IP
2. On server:

```
<pre class="code-block">sudo dnf install -y certbot python3-certbot-nginx  
sudo certbot --nginx -d app.yourdomain.com</pre>
```

3. Update `.env` :

```
<pre class="code-block">AUTH_URL=https://app.yourdomain.com  
NEXTAUTH_URL=https://app.yourdomain.com  
NEXT_PUBLIC_SITE_URL=https://app.yourdomain.com  
R2_CORS_ORIGINS=https://app.yourdomain.com,http://localhost:3000</pre>
```

4. `pm2 restart pharm-lms`
-

Part 12 — External services to update

Replace `YOUR_URL` with your IP or domain.

Service	What to configure
Google OAuth	Authorized origin: <code>YOUR_URL</code>
	Redirect: <code>YOUR_URL/api/auth/callback/google</code>
R2 CORS	Allow <code>GET</code> , <code>PUT</code> , <code>HEAD</code> from <code>YOUR_URL</code>
Paystack	Webhook: <code>YOUR_URL/api/paystack/webhook</code>
Resend	Verified domain for <code>EMAIL_FROM</code>

Troubleshooting

Problem	Fix
<code>pnpm build</code> killed	Ensure 2 GB swap is enabled (Part 4.3)
502 Bad Gateway	<code>pm2 logs pharm-lms</code> ; check app on port 3000
Auth broken	<code>AUTH_URL</code> must match what users open in the browser
DB errors	Check <code>DATABASE_URL</code> ; run <code>pnpm db:migrate:deploy</code>
Upload fails	Add site URL to R2 CORS
<code>git pull</code> fails	Check deploy key / PAT on your repo

Quick reference

Task	Command
Push code (laptop)	<code>git push origin main && git push company main</code>
Deploy (server)	<code>~/deploy-pharm-lms.sh</code>
View logs	<code>pm2 logs pharm-lms</code>
App status	<code>pm2 status</code>
Restart only	<code>pm2 restart pharm-lms</code>
Reconnect	EC2 → Connect → Instance Connect

Architecture

```
<pre class="code-block">[Laptop] git push → GitHub (Prince-213/pharm-lms)
→ GitHub (PharmAnalytics/PharmAnalytics-LMS-)
```

```
[EC2] git pull ← Prince-213/pharm-lms only
```

```
nginx :80/443 → Next.js :3000 (PM2)
```

```
↓
```

```
Neon (DATABASE_URL)
```

```
Cloudflare R2 (files)
```

```
Paystack / Resend / Google OAuth</pre>
```

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